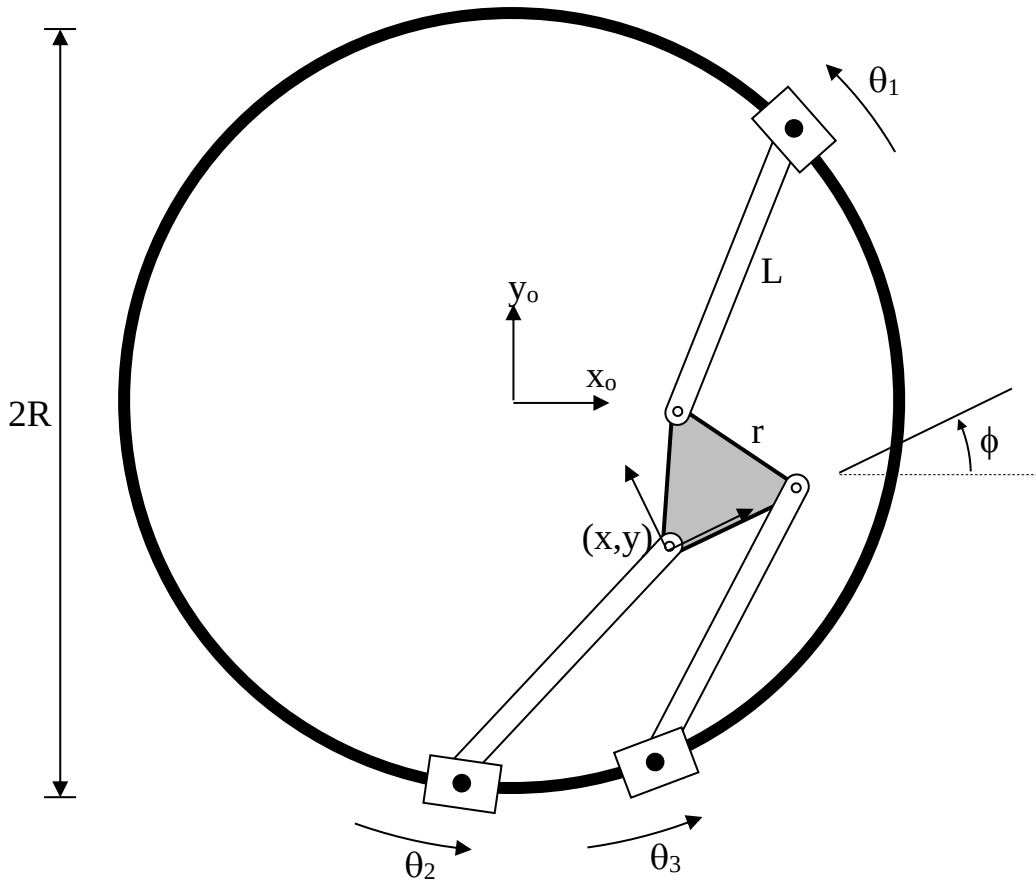




Kinematics, dynamics and control of robots - Homework project 2

The drawing depicts a 3 DOF parallel robot. The central plate is an equilateral triangle with a side length of r . The plate's vertices are connected via revolute joints to identical links of length L , whose other end is connected via revolute joints to sliders. The sliders move along a circular track of radius R . All revolute joints are passive. The robot's actuated degrees of freedom are the angular positions of the sliders along the track $\mathbf{q} = [\theta_1, \theta_2, \theta_3]^T$. The task vector is the position and orientation of the moving frame attached to the plate $\mathbf{x} = [x, y, \phi]^T$.



1. Solve the inverse kinematics of the robot, that is, find the values of the joints $\mathbf{q}$ as a function of the robot's position and orientation $\mathbf{x}$. Calculate and draw (using a computer) all the solutions of the inverse kinematics for the task $[x, y, \phi] = [1.5, 0, 10^\circ]$.
2. Solve the forward kinematics of the robot, that is, find the position and orientation of the end effector $\mathbf{x}$ as a function of the joint values $\mathbf{q}$.

Directions:

- Write kinematics constraints $\mathbf{F}(\mathbf{x}, \mathbf{q}) = \mathbf{0}$, then eliminate all $\{x^2, y^2\}$ terms to obtain a linear system in $\{x, y\}$ whose coefficients depend nonlinearly on ϕ .
- Substitute the solution for $\{x, y\}$ into one of the original equations.
- Substitute:

$$\cos \phi = \frac{1 - t^2}{1 + t^2}, \quad \sin \phi = \frac{2t}{1 + t^2}, \quad \tan \frac{\phi}{2} = t$$



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- Solve the high-degree polynomial in t numerically. Detail all the steps of the solution. There is no need to write the coefficients of the polynomial.

For joint values $\mathbf{q}^* = [90^\circ, 120^\circ, 40^\circ]^T$, calculate and draw all solutions of the forward kinematics. Write all the solutions for $[x, y, \phi]$.

Verify each solution for $\mathbf{x}$ by recalculating the inverse kinematics $\mathbf{q} = f(\mathbf{x})$. What is the error obtained? Explain.

- Plan the motion of the plate, $\mathbf{x}$ from point $\mathbf{x}_a = [1, -0.1, 0^\circ]^T$ to point $\mathbf{x}_b = [0, 0.75, 90^\circ]^T$ along a straight line at constant speed over 2 seconds, under the restriction that the links cannot cross each other. Present the analytical expression for $\mathbf{x}(t)$, plots of the joint variables' position as a function of time. In addition, draw the robot's platform and the 3 links at $t = (0, 0.5, 1, 1.5, 2)[sec]$ in the x-y plane on the same plot.
- Calculate parametrically the Jacobian matrices $J_q \dot{\mathbf{q}} = J_x \dot{\mathbf{x}}$ by differentiating the geometric constraints $\mathbf{F}(\mathbf{x}, \mathbf{q}) = 0$. Express the Jacobians J_q, J_x as a function of $\mathbf{x}, \mathbf{q}$.
- For $\phi = 10^\circ, y = 0$, and the range $0 < x < 2$, find and draw all qualitatively different, singular positions. For each position, calculate the direction of free motion where $J_x \cdot \dot{\mathbf{x}} = 0$ holds.
- Write a short paragraph summarizing the assignment, refer to the results and your conclusions.

General comment: It is recommended to use symbolic programs such as Matlab, but the printed results do not replace a detailed explanation of all the steps to the solution.

Submission in pairs by 27.5.2025 on the course site.

Robot parameters: $r = 1.5, L = 3, R = 3.5$ (all sections should be solved parametrically. Substitute values only for final numeric solutions.)

Useful Matlab commands:

Numeric calculation of polynomial roots by its coefficients	<code>sol=roots([a3 a2 a1 a0])</code>
Calculate eigenvalues and eigenvectors	<code>[M, D]=eig(A)</code>
Draw a triangle	<code>patch([x1 x2 x3], [y1 y2 y3], 'green', 'edgecolor', 'red')</code>
Allow drawing additional line on figure	<code>hold on</code>
Draw a line	<code>plot(xvec, yvec, 'blue', 'linewidth', 2)</code>
Use the same length for the data units along each axis	<code>axis equal</code>
Set figure limits	<code>axis([-1 5 0 6])</code>
Set symbolic variables	<code>syms x y</code>
Differentiate symbolic expression by x	<code>diff(f, x)</code>
Substitute values in symbolic expression	<code>subs(f, [x, y], [5, z^2])</code>
Calculate Jacobian of symbolic vector	<code>J=jacobian([Px, Py, Pz], [q1, q2, q3])</code>
Symbolic solution of system of equations	<code>solve('a*x+b*y=1', 'c*x+d*y=0', 'x, y')</code>
Simplify, reorder symbolic expression	<code>collect, simplify, factor</code>
converts a symbolic expression to a numeric	<code>matlab function</code>